PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

LendLoop – Campus Resource Sharing Application

Group members with Student Id

1. Vedant Dumbare — 24107129
2.Krutika Ghadi — 24107079
3.Tejas Shelar — 24107058

Guide
Mrs.Hardiki Achrekar

Contents

1.Introduction

2.Objectives

3.Scope

4.Features / Functionality

5.Project Outcomes

6.Technology Stack

7.Block Diagram if applicable

8.Results

9.Conclusion

10.References

SDG 12 – Responsible Consumption and Production

1. LendLoop promotes reuse of items instead of buying new one
2. Reduces waste and unnecessary consumption
3. Encourages sustainable usage of resources within campus
4. Supports efficient utilization of existing materials

1. Introduction

❑ Problem Statement

1. Students often require resources such as books, calculators, lab equipment, etc.
2. Purchasing items for short-term use can be expensive.
3. There is no organized system within the campus for sharing resources.

❑ Solution Proposed

1. Develop LendLoop, a mobile application for campus resource sharing. Students can lend, exchange, or request items from other students.
2. The platform promotes efficient resource usage and collaboration.

2. Objectives

1. To Develop a Centralized Resource Sharing Platform
2. To Develop Efficient Borrowing and Lending Process
3. To Develop Secure User Authentication
4. Real-Time Data Management

3. Scope

1. Can be implemented within college campuses
2. Useful for students who want to borrow or lend resources
3. Supports sharing of books, gadgets, and study materials
4. Can be expanded to multiple colleges in the future.

4. Feature /Functionality

- **User Login System.**
  Users can log into the application using Firebase Authentication.

- **Dashboard Interface.**
  After login, users access a dashboard to navigate through the application.

- **Item Exchange Feature.**
  Students can exchange or lend items to other students.

- **Item Search Feature.**
  Users can search available resources within the application.

5. Outcome of Project

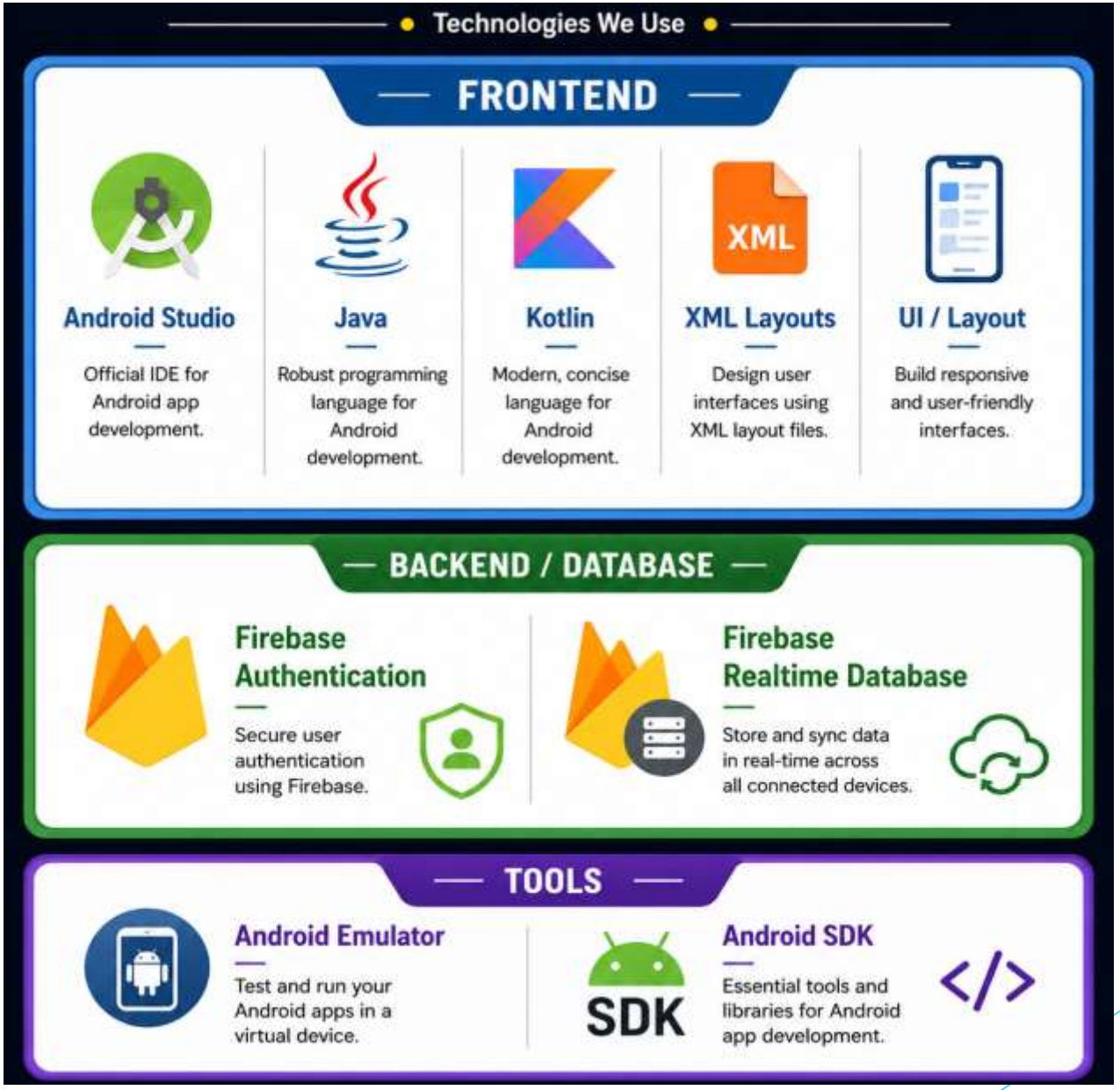
1. Users can log into the application using Firebase authentication.

2. Users can access the dashboard interface after login.

3. Users can search available items in the system. Users can exchange items with other students.

4. The application demonstrates a working prototype of campus resource sharing.

6. Technology Stack



7.Block Diagram

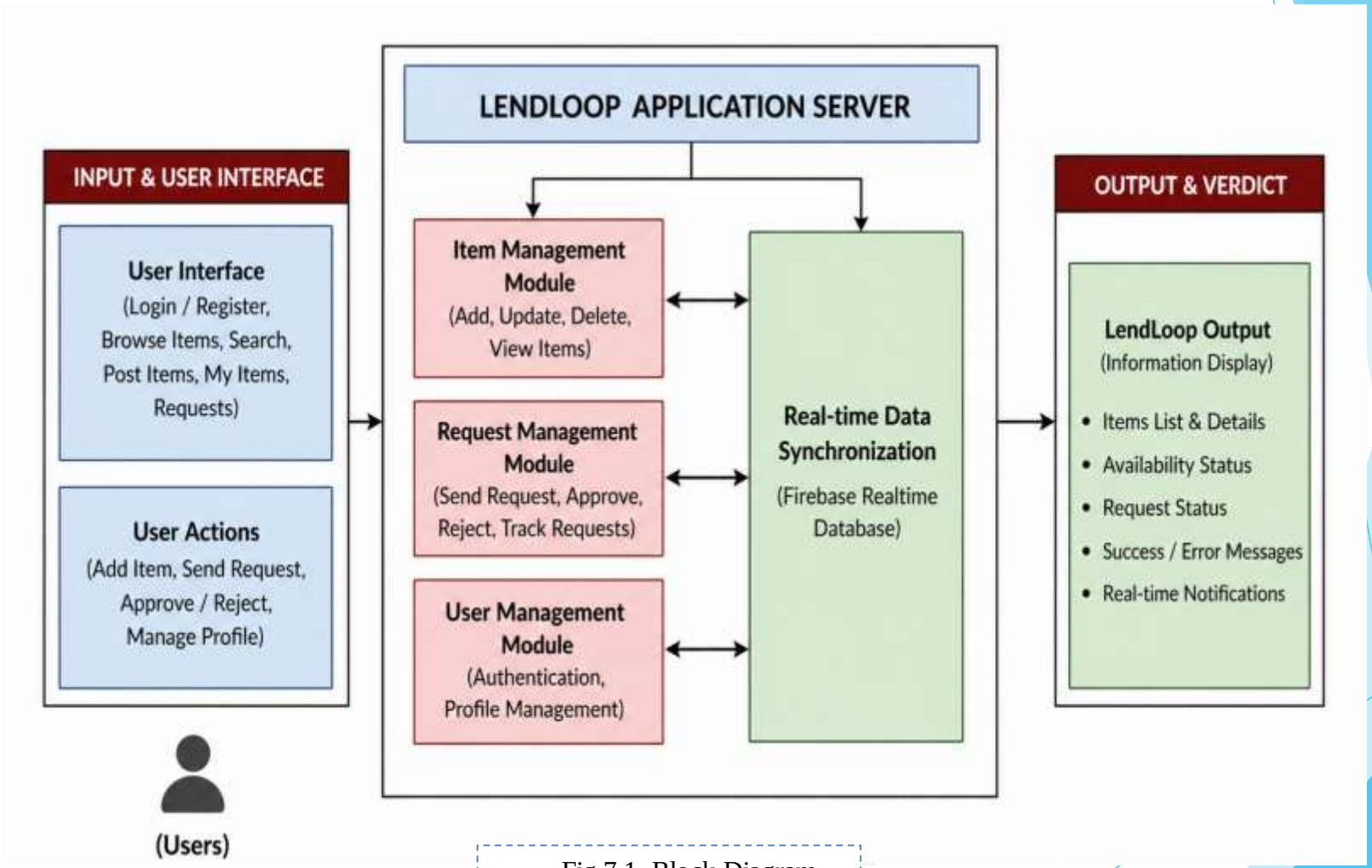


Fig.7.1 .Block Diagram

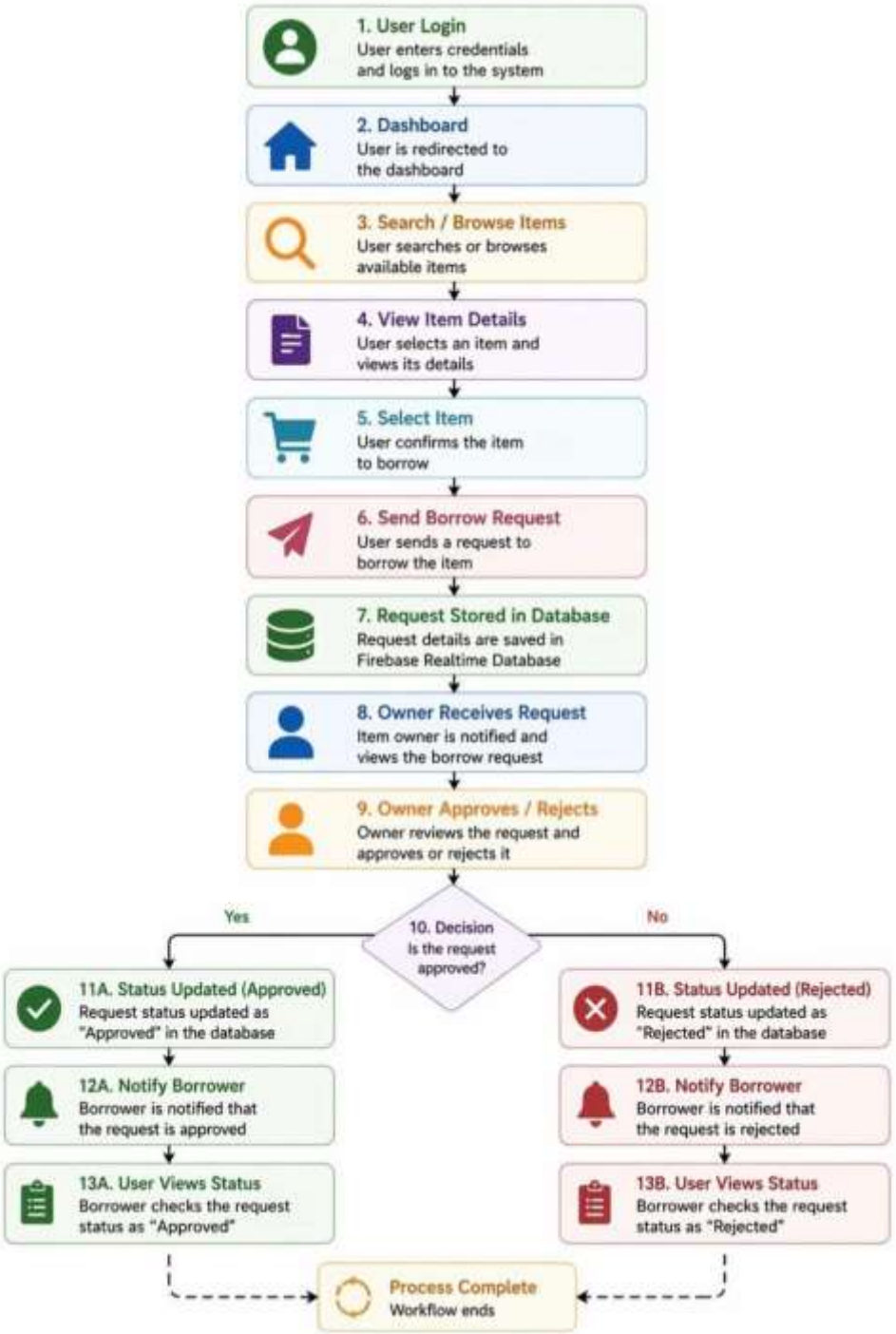


Fig.7.2 Workflow for lending an Item.

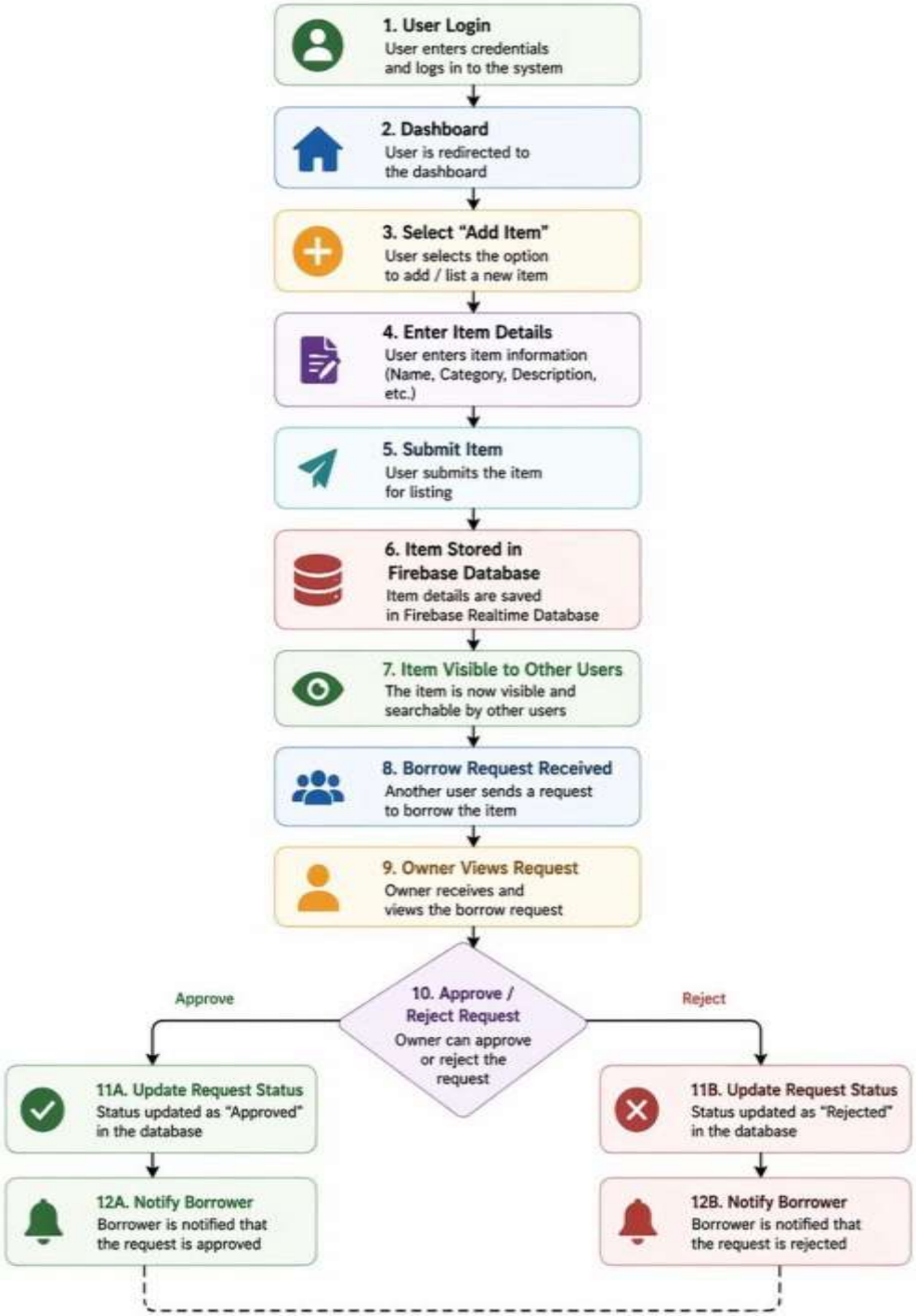
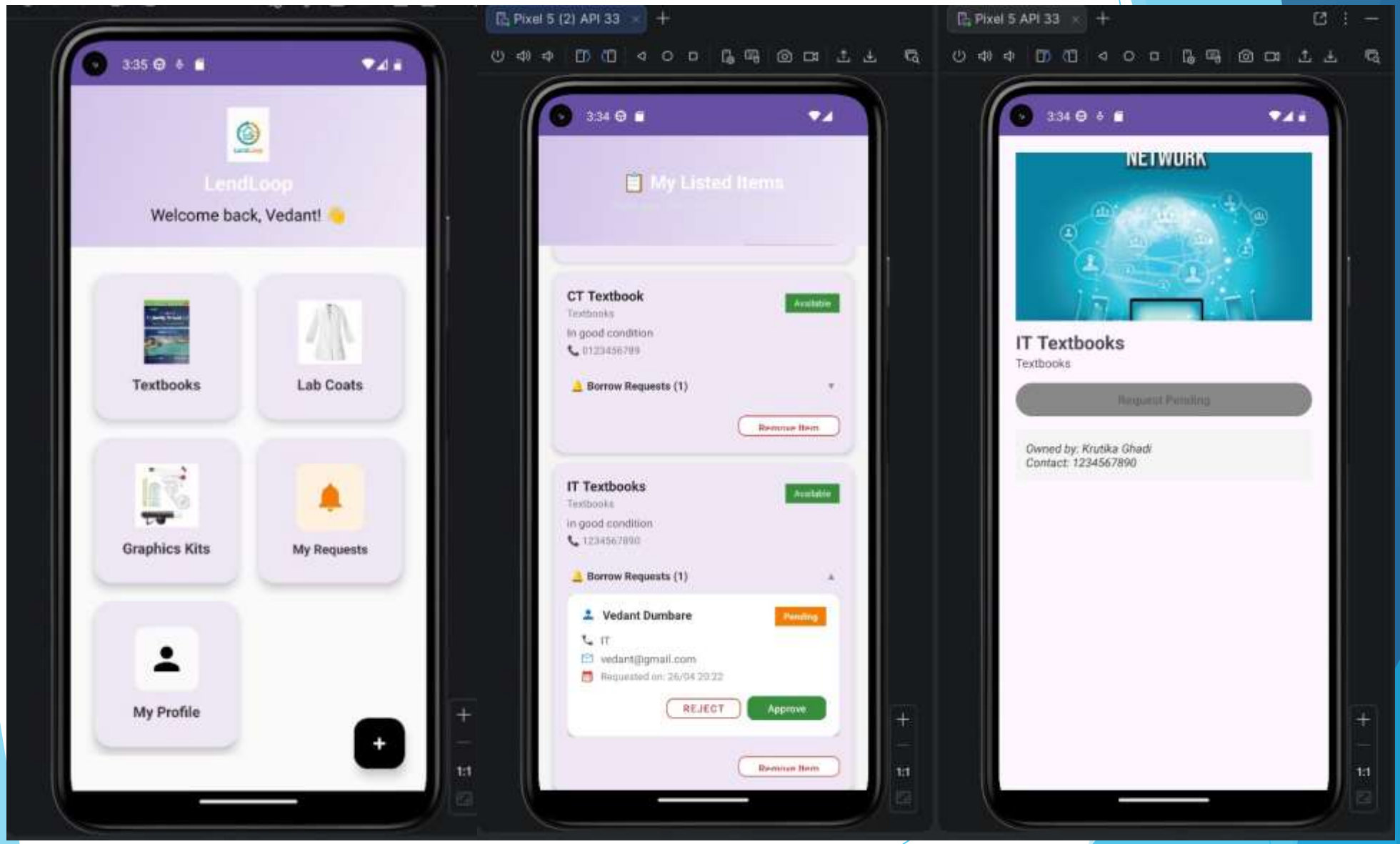


Fig.7.3 Workflow for searching and borrowing an item

8.Results


LendLoop
Welcome back, Vedant!
Textbooks
Lab Coats
Graphics Kits
My Requests
My Profile
My Listed Items
CT Textbook
Available
Textbooks
in good condition
Borrow Requests (1)
Remove Item
IT Textbooks
Available
Textbooks
in good condition
Borrow Requests (1)
Vedant Dumbare
Pending
IT
vedant@gmail.com
REJECT
Approve
Remove Item
NETWORK
IT Textbooks
Textbooks
Owned by: Krutika Ghadi
Contact: 1234567890

Comparison with Existing Systems

Criteria	Existing Systems (OLX / Rental Apps)	LendLoop (Campus-Based System)
Target Audience	Open to general public	Restricted to verified college students
Trust & Security	Low (unknown users)	High (campus-based verified users)
Primary Purpose	Buying and selling items	Borrowing and lending resources
Cost Factor	Paid transactions	Free or minimal cost
Availability	Location-dependent	Easily available within campus
Ease of Use	Time-consuming search	Quick and simple access
User Verification	Limited or no verification	Authentication using college credentials
Focus Area	Profit-oriented marketplace	Resource sharing & sustainability

9.Conclusion

1. **Centralized Resource Sharing:** LendLoop provides a structured platform for easy borrowing and lending within the campus.

2. **User-Friendly & Efficient:** Allows users to list items, search resources, and manage requests with better visibility and control.

3. **Real-Time & Secure System:** Uses Firebase for instant data updates, secure authentication, and reliable performance.

4. **Impact & Future Scope:** Reduces costs, improves collaboration, and can be enhanced with features like payments, notifications, and multi-campus support.

10.References

[1] Android Developers, Android Developer Documentation
https://developer.android.com/

[2] Firebase Documentation, Google
https://firebase.google.com/docs

[3] Firebase Authentication Documentation, Google
https://firebase.google.com/docs/auth

[4] Firebase Realtime Database Documentation, Google
https://firebase.google.com/docs/database

[5] Oracle, Java SE Documentation
https://docs.oracle.com/javase/8/docs

[6] Kotlin Programming Language Documentation
https://kotlinlang.org/docs

[7] GitHub Documentation
https://docs.github.com/

Thank You…!